Chapter 1 — Measurements

Full textbook exercise · PECTAA Physics 11 · No ALP filtering

Q.1 — Multiple Choice Questions (with correct option and reason)

1.1. The purpose of study and discoveries in Physics is:

(a) the probing of interstellar spaces
(b) the betterment of mankind
(c) the development of destructive technology in warfare
(d) development in aesthetics for the world

Answer: (b) the betterment of mankind — Physics aims to understand nature and use that knowledge for human welfare.

1.2. The length of a steel pipe is between 0.7 m and 0.8 m. The appropriate instrument for an accuracy of 0.001 m is:

(a) a micrometer screw gauge
(b) a metre rule
(c) a ten metres measuring tape
(d) a Vernier callipers

Answer: (b) a metre rule — A metre rule reads to 1 mm = 0.001 m and is suitable for this length.

1.3. The diameter of a steel ball measured by Vernier callipers as shown is:

(a) 1.30 cm
(b) 1.39 cm
(c) 1.40 cm
(d) 1.31 cm

Answer: (b) 1.39 cm — Main scale 1.3 cm plus vernier coincidence 0.09 cm gives 1.39 cm.

1.4. The micrometer screw gauge reading for the thin rod is:

(a) 3.67 mm
(b) 3.17 mm
(c) 4.17 mm
(d) 4.20 mm

Answer: (a) 3.67 mm — Sleeve 3.5 mm plus thimble 0.17 mm gives 3.67 mm.

1.5. The number of significant figures of a measurement are defined as:

(a) they reflect the accuracy of the observation
(b) they are the figures which are reasonably reliable
(c) they are the accurately known digits and the first doubtful digit
(d) all of the above

Answer: (d) all of the above — All three statements describe significant figures.

1.6. The number of significant figures in 2500.0 kg is:

(a) two
(b) three
(c) four
(d) five

Answer: (d) five — The decimal point makes the trailing zero significant.

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1.7. The sum 12 kg + 2.02 kg + 5.1 kg according to appropriate precision is:

(a) 19 kg **(b)** 19.0 kg
(c) 19.1 kg **(d)** 19.12 kg

Answer: (a) 19 kg — 12 kg has no decimal place, so the sum is rounded to 19 kg.

1.8. The answer to the subtraction (1.126 − 0.97268) to appropriate precision is:

(a) 0.15 **(b)** 0.153
(c) 0.1533 **(d)** 0.15332

Answer: (b) 0.153 — 1.126 has three decimal places, so the result is 0.153.

1.9. The product (2.8723 × 1.6) to the appropriate number of significant figures is:

(a) 4.59568 **(b)** 4.595
(c) 4.59 **(d)** 4.6

Answer: (d) 4.6 — 1.6 has two significant figures.

1.10. The division (45.2 ÷ 6.0) to appropriate significant figures is:

(a) 7.5 **(b)** 7.53
(c) 7.533 **(d)** 7.5333

Answer: (a) 7.5 — 6.0 has two significant figures.

1.11. The operation 24.4 m × 100 m / 5.0 m to appropriate significant figures is:

(a) 4680 m **(b)** 4900 m
(c) 4.88×10^3 m **(d)** 4.9×10^2 m

Answer: (d) 4.9×10^2 m — The result 488 m is rounded to two significant figures: 4.9×10^2 m.

1.12. The ratio of the dimensions of force and energy is:

(a) T **(b)** T^{-1}
(c) L **(d)** L^{-1}

Answer: (d) L^{-1} — $[MLT^{-2}]/[ML^2T^{-2}] = [L^{-1}]$.

1.13. Which pair does NOT have identical dimensions?

(a) work and torque **(b)** angular momentum and Planck's constant
(c) moment of inertia and moment of force **(d)** impulse and momentum

Answer: (c) moment of inertia and moment of force — Moment of inertia is $[ML^2]$, torque is $[ML^2T^{-2}]$.

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1.14. The length of the solid cylinder from the Vernier callipers figures is:

(a) 3.26 cm **(b)** 3.30 cm
(c) 3.34 cm **(d)** 4.20 cm

Answer: (c) 3.34 cm — The corrected vernier reading gives 3.34 cm.

1.15. The least count of an instrument determines:

(a) precision of a measurement **(b)** accuracy of a measurement
(c) fractional uncertainty of a measurement **(d)** percentage uncertainty of a measurement

Answer: (a) precision of a measurement — Least count fixes the smallest reliable division, hence precision.

1.16. A measuring tape has minimum scale division 0.2 cm. The allowed reading is:

(a) 80.5 cm **(b)** 80.6 cm
(c) 80.65 cm **(d)** 80.7 cm

Answer: (b) 80.6 cm — The reading must be consistent with 0.2 cm divisions; 80.6 cm is allowed.

Q.2 — Short Answer Questions

1.1. What are base units and derived units? Give examples.

Base units are the seven independent SI units: metre, kilogram, second, ampere, kelvin, mole and candela. Derived units are formed from base units, for example newton = kg m s^{-2}, joule = N m and watt = J s^{-1}.

1.2. How many significant figures should be retained when (i) multiplying or dividing and (ii) adding or subtracting?

(i) In multiplication or division, keep the same number of significant figures as the factor with the fewest significant figures. (ii) In addition or subtraction, keep the same number of decimal places as the term with the fewest decimal places.

1.3. How is the vernier scale related to the main scale? What is meant by L.C. of Vernier callipers?

The vernier scale has divisions slightly smaller than the main scale divisions, so a small fractional main-scale reading can be read by coincidence. The least count is the smallest length it can measure, usually 0.01 cm, equal to one main-scale division minus one vernier-scale division.

1.4. Write in scientific notation: (a) 143.7 (b) 206.4 × 10^2.

(a) 143.7 = 1.437 × 10^2. (b) 206.4 × 10^2 = 2.064 × 10^4.

1.5. Write using correct prefixes: (a) 580 × 10^{-9} g (b) 0.46 × 10^{-3} s.

(a) 580 × 10^{-9} g = 580 ng. (b) 0.46 × 10^{-3} s = 0.46 ms = 460 µs.

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1.6. Kinetic energy is ½mv². What are its dimensions?

[K] = [M][LT^{-1}]^2 = [ML^2T^{-2}]. The factor ½ is dimensionless.

1.7. How many significant figures are there in (i) 37 km (ii) 0.002953 m (iii) 7.50034 cm (iv) 200.0 m?

(i) 2, (ii) 4, (iii) 6, (iv) 4. Leading zeros are not significant; trailing zero after decimal is significant.

1.8. Write the dimensions of (i) Planck's constant (ii) angular velocity.

(i) From E = hf, [h] = [ML^2T^{-2}]/[T^{-1}] = [ML^2T^{-1}]. (ii) Angular velocity = angle/time, so [ω] = [T^{-1}].

Constructed Response Questions

1.1 Why do we find it useful to have two units for amount of substance, kilogram and mole?

The kilogram measures mass, which is convenient for weighing matter in the laboratory. The mole measures the number of particles, which is convenient in chemistry and physics because reactions and gas laws depend on particle count. One mole contains Avogadro's number of particles, so the two units connect macroscopic mass with microscopic count.

1.2 Three students measured a rod with a scale of minimum division 1 mm and recorded 0.4235 m, 0.42 m and 0.424 m. Which is correct and why?

The correct record is 0.424 m. A 1 mm scale can reliably read to 0.001 m, so the reading should have three decimal places in metres. 0.4235 m claims too much precision, while 0.42 m gives too little.

1.3 Why is the kilogram, not the gram, the base unit of mass?

The SI base unit was chosen as the kilogram for historical and practical reasons: it is a convenient size for laboratory standards and coherent derived units such as the newton and joule. The gram is a submultiple, 10^{-3} kg.

1.4 If P = Q + R and Q, R both have dimensions [MLT], what are dimensions and SI unit of P? If Q and R had different dimensions, could P be determined?

Only quantities with the same dimensions can be added, so P has dimensions [MLT] and SI unit kg m s^{-1}. If Q and R had different dimensions, the equation would be invalid and P could not be determined.

1.5 What is the least count of a clock with (a) hour, minute and second hands (b) hour and minute hands?

(a) With a second hand, the least count is 1 s. (b) With only hour and minute hands, the smallest readable division is 1 min.

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1.6 How can the diameter of a round pencil be measured with a metre rule as accurately as with Vernier callipers?

Wind a thread closely around the pencil for a known number of turns, measure the total length of the thread with the metre rule, and divide by the number of turns to get the circumference. Then d = circumference/π. Measuring many turns reduces the percentage uncertainty.

1.7 How would readings differ if a screw gauge is used instead of Vernier callipers to measure a glass plate?

A screw gauge has a smaller least count, usually 0.001 cm, compared with 0.01 cm for Vernier callipers. It therefore gives a more precise reading and smaller absolute uncertainty for a thin glass plate.

1.8 Write the correct reading of the cylinder if the Vernier callipers has an error of 0.02 cm.

Correct reading = observed reading − zero error. If the observed reading is 3.34 cm and the zero error is +0.02 cm, the correct length is 3.34 − 0.02 = 3.32 cm.

1.9 A screw gauge has 50 circular divisions; 10 revolutions advance 5 mm. The 2nd division coincides with the datum line when closed. Find the thickness shown.

Pitch = 5 mm/10 = 0.5 mm. Least count = 0.5/50 = 0.01 mm. Zero error = +2 × 0.01 = +0.02 mm. Correct thickness = main scale + circular scale × LC − zero error, using the figure reading.

1.10 What is meant by a dimensionless quantity? Give one example.

A dimensionless quantity has no base dimensions; it is a ratio of similar quantities. Example: strain = extension/original length, or angle in radians = arc/radius.

1.11 A student folds a paper three times and measures total thickness with a screw gauge. How is the sheet thickness found?

Three folds give $2^3 = 8$ layers. The thickness of one sheet is the measured total thickness divided by 8, after correcting for zero error.

1.12 Round to 3 significant figures and write in scientific notation: (a) 0.02055 (b) 4656.5.

(a) $0.02055 \rightarrow 2.06 \times 10^{-2}$. (b) $4656.5 \rightarrow 4.66 \times 10^{3}$.

Comprehensive / Long Questions

1.1 What is meant by uncertainty in a measurement? How is uncertainty in a digital instrument indicated?

Uncertainty is the estimated range within which the true value is expected to lie. It arises because every instrument has a finite least count and because random errors affect readings. For a digital instrument, the last digit may fluctuate; if it fluctuates by one or two steps, that digit is still recorded and the uncertainty is taken as the smallest digital step or the observed fluctuation.

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1.2 Differentiate between precision and accuracy with reference to measurement.

Precision is the closeness of repeated readings to one another and depends on the least count; it is expressed by absolute uncertainty. Accuracy is the closeness of a reading to the true value and is expressed by fractional or percentage uncertainty. A fine instrument can be precise but inaccurate if it has a zero error.

Feature	Precision	Accuracy
Meaning	Repeatability of readings	Closeness to true value
Depends on	Least count	Fractional/percentage uncertainty
Improved by	Finer instrument	Removing systematic error

1.3 (a) What are significant figures? Give two reasons for using them. How is uncertainty found in a timing experiment such as a pendulum? (b) A cylinder has mass 12.85 g, length 3.35 cm and diameter 1.25 cm. Find density with uncertainty.

(a) Significant figures are the reliably known digits plus the first doubtful digit. They show the precision of a measurement and prevent false accuracy in calculations. In a pendulum experiment, time for many oscillations is measured and divided by the number of oscillations; the uncertainty in the period is the timing uncertainty divided by the number of oscillations. (b) Volume $V = \pi d^2 L/4 = 4.12\ \text{cm}^3$. Density $\rho = m/V = 3.12\ \text{g cm}^{-3}$. Percentage uncertainty $= (\Delta m/m) + 2(\Delta d/d) + (\Delta L/L)$; using least counts gives about 1.8%, so $\rho \approx 3.12 \pm 0.06\ \text{g cm}^{-3}$.

1.4 Explain with examples how physical quantities are written in dimensions. Give two benefits.

Dimensions express a quantity in terms of base quantities [M], [L], [T]. For example, force = $[MLT^{-2}]$, energy = $[ML^2T^{-2}]$, power = $[ML^2T^{-3}]$. Benefits: (1) checking the homogeneity of equations; (2) deriving simple relations and converting units.

1.5 Check the homogeneity of $v = \sqrt{(T/\mu)}$, where v is wave speed on a string, T tension and μ mass per unit length.

$[T] = [MLT^{-2}]$, $[\mu] = [ML^{-1}]$. So $[T/\mu] = [L^2T^{-2}]$ and $\sqrt{[T/\mu]} = [LT^{-1}]$, which is the dimension of speed. The equation is dimensionally homogeneous.

Numerical Problems (fully worked)

1.1 If speed of light is 3×10^8 m s⁻¹, what is one light year in metres?

Given: $c = 3 \times 10^8\ \text{m s}^{-1}$, 1 year $\approx 3.156 \times 10^7$ s

Formula: distance = speed × time

1 ly $= 3 \times 10^8 \times 3.156 \times 10^7 \approx 9.5 \times 10^{15}$ m.

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Answer: 9.5×10^{15} m

1.2 Write estimated answers in standard form: (a) seconds in one year (b) years in one second.

(a) 1 year ≈ 3.2×10^{7} s. (b) 1 s ≈ 3.1×10^{-8} year.

Answer: (a) 3.2×10^{7} s (b) 3.1×10^{-8} year

1.3 Length and width of a plate are 18.3 cm and 14.60 cm. Find area to correct significant figures.

Given: L = 18.3 cm (3 s.f.), W = 14.60 cm (4 s.f.)

Formula: A = L × W

A = 18.3 × 14.60 = 267.18 cm^2. Round to 3 significant figures.

Answer: 267 cm^2

1.4 Find the sum of the given masses to appropriate precision.

Adding the masses and rounding to the least precise decimal place gives 20.4 kg, as printed in the book.

Answer: 20.4 kg

The OCR of the individual masses is unclear; the book answer is retained.

1.5 A cylinder measured with Vernier callipers of L.C. 0.01 cm has diameter 1.22 cm and length 5.35 cm. Find volume and uncertainty.

Given: d = 1.22 ± 0.01 cm, L = 5.35 ± 0.01 cm

Formula: $V = \pi d^2 L/4$

V = 3.1416 × $(1.22)^2$ × 5.35 / 4 ≈ 6.24 cm^3.

Percentage uncertainty = 2(0.01/1.22)×100 + (0.01/5.35)×100 ≈ 1.8%.

Absolute uncertainty ≈ 0.11 cm^3, rounded to 0.1 cm^3.

Answer: 6.24 ± 0.1 cm^3

1.6 Show that $v_f^2 - v_i^2 = 2aS$ is dimensionally correct.

$[v^2] = [L^2T^{-2}]$. $[aS] = [LT^{-2}][L] = [L^2T^{-2}]$. Both sides have the same dimensions.

Answer: Dimensionally correct

1.7 Show that $E = mc^2$ is dimensionally consistent.

$[E] = [ML^2T^{-2}]$. $[mc^2] = [M][LT^{-1}]^2 = [ML^2T^{-2}]$.

Answer: Dimensionally consistent

1.8 Derive the time period of a simple pendulum using dimensional analysis.

Given: T may depend on length ℓ, mass m, angle θ and g

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Formula: T ∝ ℓ^a m^b g^c

Angle is dimensionless. Equating dimensions: [T] = [L]^a [M]^b [LT^{-2}]^c gives a = ½, b = 0, c = −½.

Therefore T = constant × √(ℓ/g). The constant cannot be found by dimensions; experiment gives 2π.

Answer: T = constant √(ℓ/g)

Textbook checks flagged in this chapter

• The book prints two questions numbered 1.16 in the MCQ section; the second is the measuring-tape question.

• Figure-based readings (1.3, 1.4, 1.14, 1.8, 1.9, 1.11) depend on the printed diagrams; the readings used here follow the textbook figures.

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